

Black Hole Horizons and Hawking Temperature as Projection Phenomena in the Methane Metauniverse

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Abstract

We derive black-hole horizon structure, Hawking temperature, and entropy within the Methane Metauniverse (MMU) framework without invoking quantum field theory in curved spacetime. Gravitation emerges as elastic compression of a discrete tetrahedral spacetime lattice, while observable time arises from projection onto a distinguished internal axis. We show that the Schwarzschild metric component g_{00} follows inevitably from the MMU Lagrangian via an energy-closure condition. The Hawking temperature and Bekenstein–Hawking entropy then arise as direct consequences of projection gradients and microscopic state counting at the Planck fixed point.

1 MMU Field Equation and Spatial Deformation

In the MMU spacetime model, each ur-cell possesses an internal edge length a . Gravitational fields deform this length according to

$$a(r) = a_0 [1 + u(r)]. \quad (1)$$

In the weak-field continuum limit the MMU elastic field equation reads

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = \frac{4\pi G}{c^2} \rho_{\text{eff}}(r). \quad (2)$$

Outside the mass distribution ($\rho_{\text{eff}} = 0$) this yields

$$u(r) = -\frac{GM}{rc^2}. \quad (3)$$

2 Projection Origin of Time Dilation

Observable time t is defined as projection of internal motion onto the distinguished axis w_1 . For an internal velocity $\vec{v} = (v_1, \vec{v}_\perp)$ the MMU projection theorem gives

$$\frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v_\perp^2}{v_{\text{int}}^2}}}. \quad (4)$$

Hence

$$\frac{w_1(r)}{w_1(\infty)} = \sqrt{1 - \frac{v_\perp^2}{c^2}}. \quad (5)$$

3 Energy Closure and Unavoidable Resummation

The internal MMU Lagrangian for one ur-cell is

$$L = \frac{1}{2} \dot{w}^T M \dot{w} - \frac{1}{2} w^T K(a) w, \quad (6)$$

with stiffness $K(a) \propto a^{-3}$. In a static gravitational field $a(r)$ varies adiabatically, so total internal energy is conserved:

$$dE = dT + dV = 0. \quad (7)$$

Compression energy is stored exclusively in transverse modes, implying

$$\frac{v_{\perp}^2}{2} = -\Phi(r), \quad \Phi(r) = c^2 u(r). \quad (8)$$

Substitution yields

$$\frac{v_{\perp}^2}{c^2} = \frac{2GM}{rc^2}. \quad (9)$$

Therefore the projection factor becomes

$$\frac{w_1(r)}{w_1(\infty)} = \sqrt{1 - \frac{2GM}{rc^2}}. \quad (10)$$

This implies

$$g_{00}(r) = \left(\frac{w_1(r)}{w_1(\infty)} \right)^2 = 1 - \frac{2GM}{rc^2}, \quad (11)$$

showing that the Schwarzschild form arises *inevitably* from MMU dynamics without additional assumptions.

4 Horizon and Hawking Temperature

The horizon corresponds to vanishing projection, $w_1(r_s) = 0$, yielding

$$r_s = \frac{2GM}{c^2}. \quad (12)$$

The surface gravity follows from the projection gradient,

$$\kappa = \frac{c^2}{2} \left. \frac{dg_{00}}{dr} \right|_{r=r_s} = \frac{c^4}{4GM}. \quad (13)$$

The MMU Hawking temperature is therefore

$$T_{\text{MMU}} = \frac{\hbar \kappa}{2\pi k_B c} = \frac{\hbar c^3}{8\pi G M k_B}, \quad (14)$$

identical to Hawking's result but obtained without quantum fields in curved spacetime.

5 Entropy from Microscopic Cell States

At the horizon the MMU lattice reaches the Planck fixed point, $a = \ell_{\text{Pl}}$. The number of available ur-cells on the horizon is

$$N = \frac{A}{\ell_{\text{Pl}}^2}. \quad (15)$$

Counting independent projection states yields

$$S_{\text{MMU}} = \frac{k_B}{4} \frac{A}{\ell_{\text{Pl}}^2}, \quad (16)$$

recovering the Bekenstein–Hawking entropy.

6 Conclusion

Black holes are fully compatible with the MMU framework. Horizons correspond to projection collapse rather than spacetime singularities, while Hawking temperature and entropy emerge from elastic deformation and microscopic state counting. The derivation requires no external general-relativistic or quantum-field postulates, demonstrating that the MMU survives the strongest consistency test of modern theoretical physics.